AI AUTOMATION PORTFOLIO PROJECT

LangGraph Multi-Agent Support System

Coordinator-worker orchestration with RAG, LLMOps observability and intelligent human escalation

LangGraph | LangChain | Claude AI | Qdrant | LangFuse | Python

Gayatri Anne

AI & Cloud Architect

github.com/gayatrianne

From Rule-Based Routing to Intelligent Agent Orchestration

The Problem

Customer support teams waste time manually triaging requests. Rule-based routing systems require hardcoded logic for every scenario and cannot handle ambiguous or novel requests gracefully.

The Solution

A multi-agent AI system where a coordinator agent classifies customer intent at runtime and dynamically delegates to specialist worker agents — billing, technical, retention, or knowledge base — with automatic human escalation for unclassifiable requests.

Key Benefits

- Dynamic intent routing
- RAG-grounded responses
- Human escalation
- Full audit trail

6	2	5	Top-3
Specialist agents in the system	LLM tiers — Haiku and Sonnet	Intent categories handled	Knowledge chunks retrieved per query

02 / HOW IT WORKS

Agent Execution Flow

1. Customer message
Received via CLI or API endpoint.

2. Coordinator agent
Classifies intent using Claude Haiku (temp=0, deterministic).

3. Conditional edge
Router function evaluates state at runtime and selects the worker.

4. Worker agent
Specialist agent resolves the query — billing / technical / retention / KB.

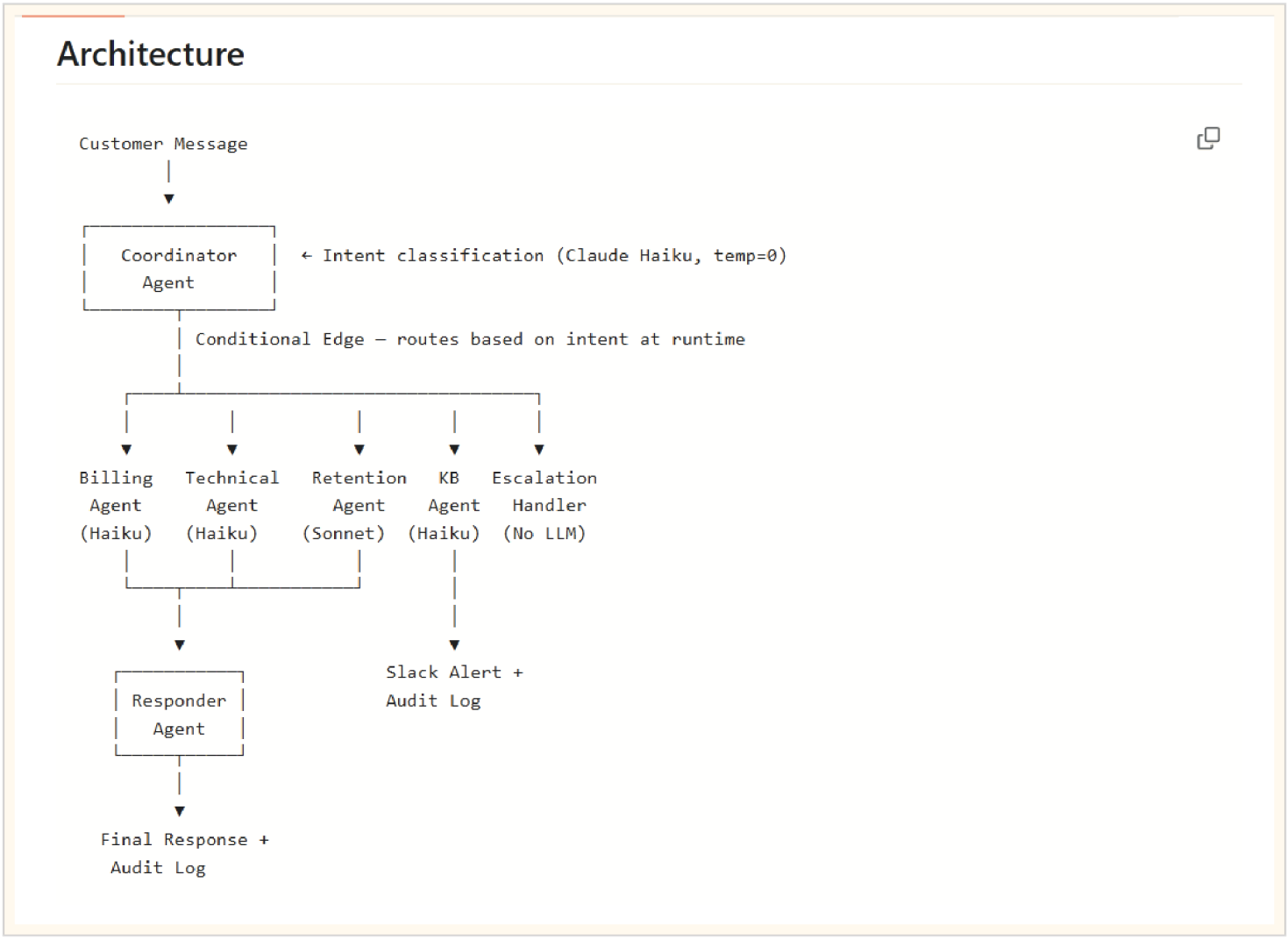
5. KB agent (if routed)
Embeds the query, searches Qdrant, retrieves top-3 chunks, grounds the response in retrieved context.

6. Responder agent
Polishes the worker result into a branded Galaxy Telecom response.

7. Audit log
SQLite record written for every interaction.

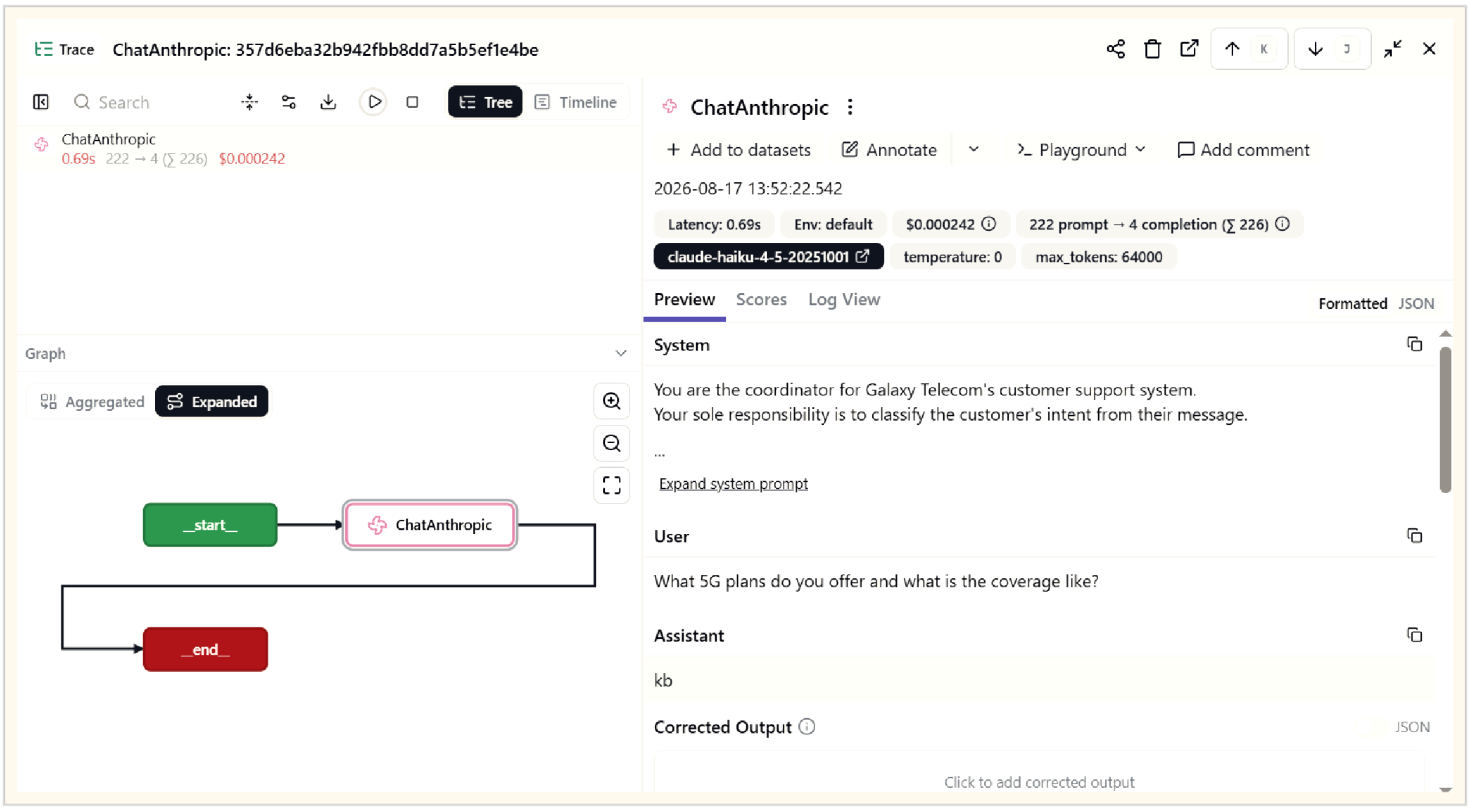
8. Escalation path
Unclassifiable intent triggers a Slack alert to the human agent team.

System Architecture



LangGraph StateGraph — coordinator with conditional edges to specialist worker agents

Execution Trace



LangFuse — one trace expanded: model, temperature, latency and token counts alongside the executed graph

Tiered Model Design

> *Not every agent requires the same model. Task complexity and business impact determine the right LLM for each role.*

AGENT	MODEL	TEMP.	RATIONALE
Coordinator	Claude Haiku	0	Deterministic single-word classification
Billing	Claude Haiku	0.3	Structured, accurate responses
Technical	Claude Haiku	0.3	Step-by-step guidance, constrained output
KB	Claude Haiku	0.3	RAG-grounded — context does the heavy lifting
Retention	Claude Sonnet	0.4	Empathetic persuasion — quality affects revenue
Responder	Claude Haiku	0.3	Brand tone and response polish

Cost note

5 of 6 agents run on Haiku — approximately 80% token cost reduction versus all-Sonnet, with quality maintained where it matters most.

Billing Intent

```
============================================================
GALAXY TELECOM — MULTI-AGENT SUPPORT SYSTEM
============================================================
Customer: My bill this month is much higher than usual, can you help?
------------------------------------------------------------
[Coordinator] Intent classified as: 'billing'
[Router] Routing intent 'billing' to 'billing_node'
[Billing Agent] Processing query: 'My bill this month is much higher than usual, can you help?...'
[Billing Agent] Resolution generated.
[Responder] Assembling final response for intent: 'billing'
[Responder] Final response assembled.
[Responder] Audit log entry written.
------------------------------------------------------------
Response:
Dear Valued Customer,

Thank you for reaching out to Galaxy Telecom. We're happy to help you understand your bill this month.

To investigate your higher charges, we'll need a few details:

- What services do you have with us? (mobile, broadband, landline, etc.)
- How much higher is this bill compared to your usual amount?
- Have there been any recent changes? (new services, usage patterns, or plan upgrades)
- Do you have any international calls or roaming charges?

**Common reasons for bill increases include:**
- Out-of-bundle usage charges
- New services or add-ons
- Plan changes
- One-off charges

We recommend checking your itemized bill in your online account for a detailed breakdown of charges. If you'd like our billing team to review this with you, please contact them:

📧 **Email:** billing@galaxy-telecom.co.uk
📞 **Phone:** Monday-Friday, 9am-6pm GMT

They'll investigate your account and explain any unexpected charges.

We're here to help—please don't hesitate to reach out if you have any other questions.

Best regards,

**Galaxy Telecom Customer Support Team**
============================================================
```

Billing agent — structured guidance with contact details

Technical Intent

```
============================================================
GALAXY TELECOM — MULTI-AGENT SUPPORT SYSTEM
============================================================
Customer: My broadband has been down since yesterday morning.
------------------------------------------------------------
[Coordinator] Intent classified as: 'technical'
[Router] Routing intent 'technical' to 'technical_node'
[Technical Agent] Processing query: 'My broadband has been down since yesterday morning....'
[Technical Agent] Resolution generated.
[Responder] Assembling final response for intent: 'technical'
[Responder] Final response assembled.
[Responder] Audit log entry written.
------------------------------------------------------------
Response:
**Subject: We're Here to Help Get Your Broadband Back Online**

Dear Valued Customer,

Thank you for contacting Galaxy Telecom. I'm sorry to hear your broadband has been down since yesterday – I understand how frustrating that must be.

Let me help you get back online. Please try these quick steps:

1. **Restart your router** – Unplug it for 30 seconds, then plug it back in and wait 2-3 minutes to fully restart
2. **Check all cables** – Ensure your power and broadband cables are firmly connected
3. **Check for outages** – Visit status.galaxy-telecom.co.uk to see if there are any known issues in your area

If these steps don't work, please let me know:
- Which lights are showing on your router?
- Are you seeing any error messages?

If the issue persists, our technical team is ready to help. Contact us at **technical@galaxy-telecom.co.uk** or call during business hours, and we'll arrange the support you need – whether
that's an engineer visit or further investigation.

We're here for you 24/7.

Warm regards,

**Galaxy Telecom Customer Support Team**
============================================================
```

Technical agent — step-by-step troubleshooting

Retention Intent — Claude Sonnet

```
============================================================
GALAXY TELECOM — MULTI-AGENT SUPPORT SYSTEM
============================================================
Customer: I am thinking about cancelling my contract with Galaxy Telecom.
------------------------------------------------------------
[Coordinator] Intent classified as: 'retention'
[Router] Routing intent 'retention' to 'retention_node'
[Retention Agent] Processing query: 'I am thinking about cancelling my contract with Galaxy Telec...'
[Retention Agent] Resolution generated.
[Responder] Assembling final response for intent: 'retention'
[Responder] Final response assembled.
[Responder] Audit log entry written.
------------------------------------------------------------
Response:
Dear Valued Customer,

Thank you for reaching out to us before making your decision — we truly appreciate the opportunity to help.

We're sorry to hear you're considering leaving Galaxy Telecom. To better assist you, could you share what's prompted this decision? Whether it's a service concern, pricing, or something els
e, we'd like to fully understand your situation.

Depending on your tenure with us, you may qualify for loyalty benefits. For example, customers with us for over 2 years are eligible for **20% off their plan** — which could make a meaningf
ul difference.

We're committed to finding a genuine solution that works for you. Please let us know what's been going on, and we'll work together to address your concerns.

We look forward to hearing from you.

Warm regards,

**Galaxy Telecom Customer Support Team**
============================================================
```

Retention agent (Claude Sonnet) — empathetic, prose-first response

Knowledge Base Intent — RAG

```
============================================================
GALAXY TELECOM — MULTI-AGENT SUPPORT SYSTEM
============================================================
Customer: What 5G plans do you offer and what is the coverage like?
------------------------------------------------------------
[Coordinator] Intent classified as: 'kb'
[Router] Routing intent 'kb' to 'kb_node'
[KB Agent] Processing query: 'What 5G plans do you offer and what is the coverage like?...'
[KB Agent] Retrieved 3 chunks from Qdrant.
[KB Agent] Resolution generated.
[Responder] Assembling final response for intent: 'kb'
[Responder] Final response assembled.
[Responder] Audit log entry written.
------------------------------------------------------------
Response:
Dear Valued Customer,

Thank you for your interest in Galaxy Telecom's 5G plans!

We're pleased to offer three mobile plans designed to suit different needs: **StarterSIM**, **ConnectPlus**, and **EliteMax**.

I can share details about our **ConnectPlus** plan at £25/month, which features 30GB of data, unlimited UK and EU calls and texts, Wi-Fi calling, and 24/7 live chat support.

For specific information about 5G access across our plans and detailed coverage details in your area, I'd recommend reaching out to our specialist team at **info@galaxy-telecom.co.uk**. The
y'll be able to provide comprehensive coverage maps and help you select the perfect plan for your needs.

We're here to help! Please don't hesitate to contact us if you have any other questions.

Best regards,

**Galaxy Telecom Customer Support Team**
============================================================
```

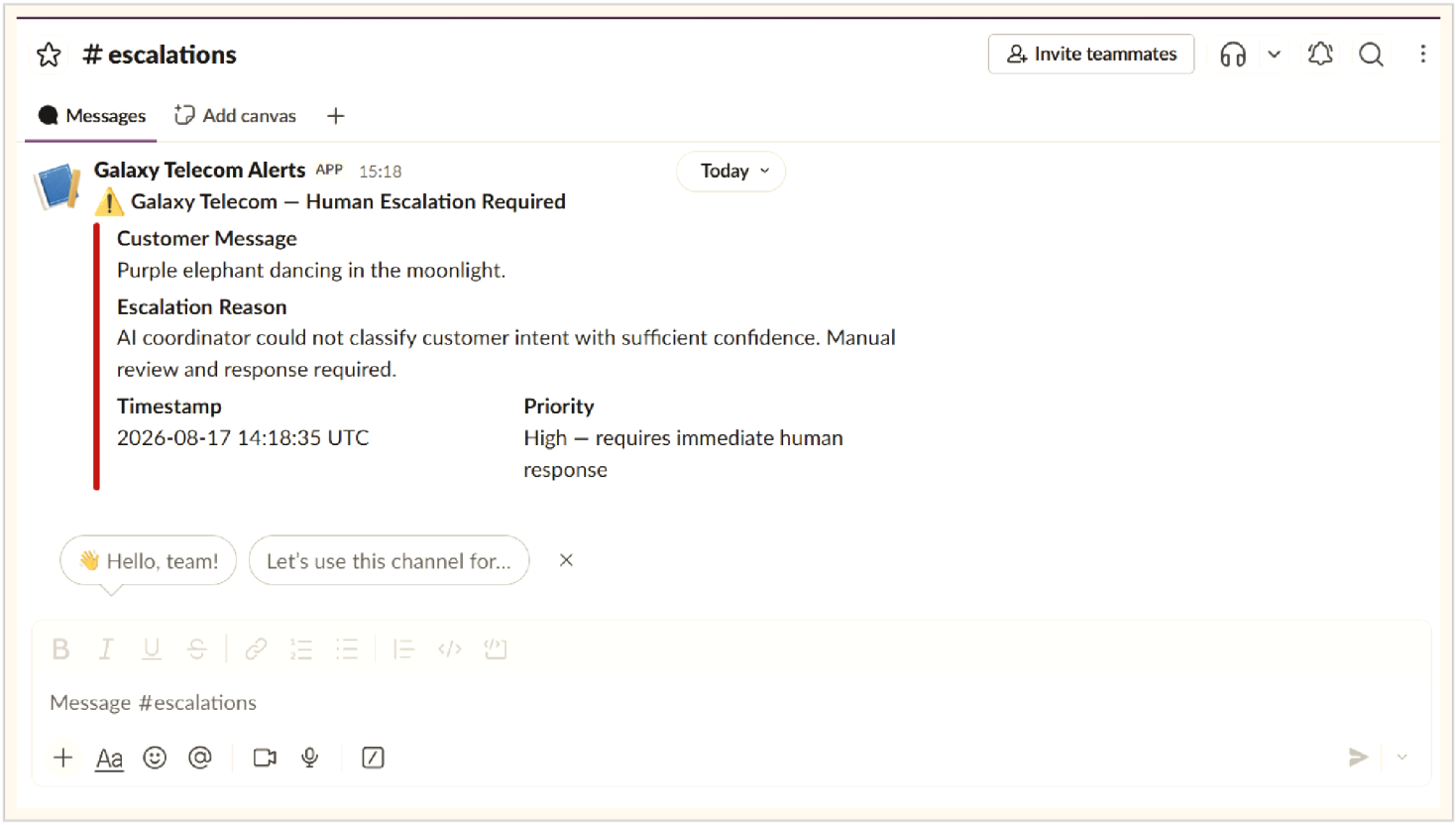
KB agent — three chunks retrieved from Qdrant, response grounded in retrieved context

Escalation Path

```
============================================================
GALAXY TELECOM — MULTI-AGENT SUPPORT SYSTEM
============================================================
Customer: Purple elephant dancing in the moonlight.
------------------------------------------------------------
[Coordinator] Intent classified as: 'unknown'
[Router] Routing intent 'unknown' to 'escalation_node'
[Escalation Handler] Escalating request to human agent team.
[Escalation Handler] Slack alert sent successfully.
[Escalation Handler] Audit log entry written.
------------------------------------------------------------
Response:
Thank you for contacting Galaxy Telecom. We want to make sure your query receives the attention it deserves. One of our specialist agents will review your message and get back to you within
 2 hours. We apologise for any inconvenience.
============================================================
```

Escalation handler — unclassifiable intent routed away from the worker agents

Human Escalation Alert



Slack alert to the human agent team — customer message, reason, timestamp and priority

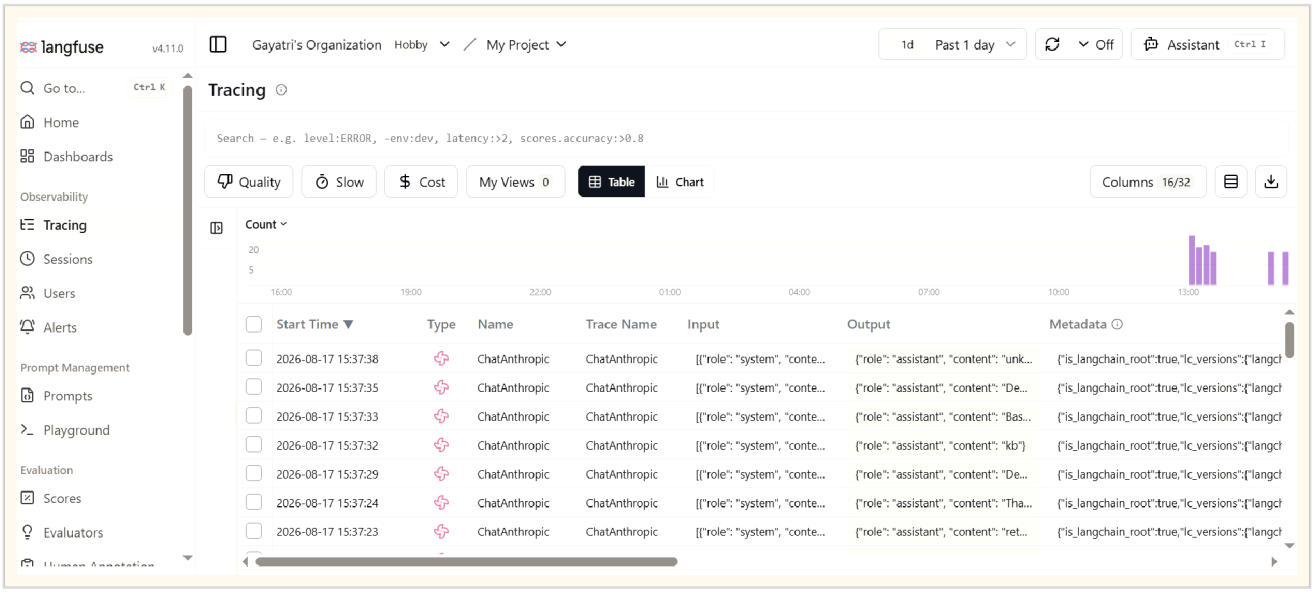
Dual-Layer Observability

LangFuse — LLM level

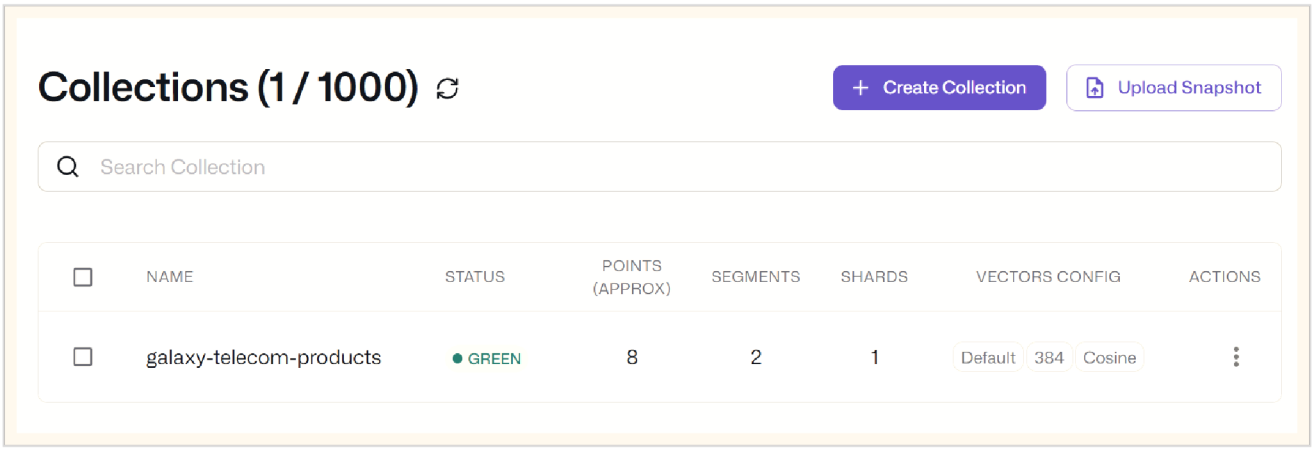
Every LLM call traced with token counts, latency per node, and the full execution graph. Accessed via cloud.langfuse.com.

SQLite audit log — business level

Structured record of every interaction — intent, worker used, resolution, escalation flag and timestamp. Queryable for reporting and analysis.



LangFuse trace list — every agent call captured with timing and token metadata



Qdrant Cloud — the galaxy-telecom-products collection backing the KB agent

Audit Log

```
(venv) C:\Gayatri\AI Demo\langgraph-multi-agent-support>python -c "from audit.logger import read_logs; import json; [print(json.dumps(log, indent=2)) for log in read_logs()[:3]]"
{
  "id": 38,
  "timestamp": "2026-08-17T14:37:39.362531",
  "customer_message": "Purple elephant dancing in the moonlight.",
  "intent": "unknown",
  "worker_used": "escalation_handler",
  "worker_result": "Escalated to human agent",
  "final_response": "Thank you for contacting Galaxy Telecom. We want to make sure your query receives the attention it deserves. One of our specialist agents will review your message and g
et back to you within 2 hours. We apologise for any inconvenience.",
  "escalated": 1,
  "escalation_reason": "AI coordinator could not classify customer intent with sufficient confidence. Manual review and response required."
}
{
  "id": 37,
  "timestamp": "2026-08-17T14:37:38.067806",
  "customer_message": "What 5G plans do you offer and what is the coverage like?",
  "intent": "kb",
  "worker_used": "kb",
  "worker_result": "Based on the information available, I can tell you that Galaxy Telecom offers three mobile plans: **StarterSIM**, **ConnectPlus**, and **EliteMax**.\n\nI have details ab
out **ConnectPlus** (\u00a325/month), which includes:\n- 30GB of data\n- Unlimited UK and EU calls and texts\n- Wi-Fi calling\n- 24/7 live chat support\n\nHowever, the excerpts provided don
't contain specific information about:\n- Which plans include 5G access\n- 5G coverage details\n- Full details of StarterSIM and EliteMax plans\n\nFor comprehensive information about 5G ava
ilability across our plans and detailed coverage maps, please contact Galaxy Telecom directly at **info@galaxy-telecom.co.uk**, and our team will be happy to help you find the right plan fo
r your needs.",
  "final_response": "Dear Valued Customer,\n\nThank you for your interest in Galaxy Telecom's 5G plans!\n\nWe're pleased to offer three mobile plans designed to suit different needs: **Star
terSIM**, **ConnectPlus**, and **EliteMax**. \n\nI can share details about our **ConnectPlus** plan at \u00a325/month, which features 30GB of data, unlimited UK and EU calls and texts, Wi-F
i calling, and 24/7 live chat support.\n\nFor specific information about 5G access across our plans and detailed coverage details in your area, I'd recommend reaching out to our specialist
team at **info@galaxy-telecom.co.uk**. They'll be able to provide comprehensive coverage maps and help you select the perfect plan for your needs.\n\nWe're here to help! Please don't hesita
te to contact us if you have any other questions.\n\nBest regards,\n\n**Galaxy Telecom Customer Support Team**",
  "escalated": 0,
  "escalation_reason": ""
}
{
  "id": 36,
  "timestamp": "2026-08-17T14:37:32.189447",
  "customer_message": "I am thinking about cancelling my contract with Galaxy Telecom.",
  "intent": "retention",
  "worker_used": "retention",
  "worker_result": "Thank you for reaching out before making your decision \u2014 I really appreciate that, and I'd love the chance to help.\n\nI'm sorry to hear you're considering leaving
us. Could you share a little more about what's prompted this? Whether it's a service issue, pricing concerns, or something else entirely, I want to make sure we fully understand your situat
ion before you make any decisions.\n\nDepending on how long you've been with us, you may also be eligible for some loyalty benefits \u2014 for example, customers with us for over 2 years qu
```

SQLite audit log — intent, worker used, resolution, escalation flag and reason per interaction

07 / TECH STACK

Components

Every component chosen for portfolio relevance and minimal infrastructure cost.

COMPONENT	TECHNOLOGY
Agent orchestration	LangGraph 1.x
LLM framework	LangChain
LLM provider	Anthropic Claude API (Haiku + Sonnet)
Vector database	Qdrant Cloud (free tier)
Embedding model	sentence-transformers all-MiniLM-L6-v2
LLM observability	LangFuse Cloud (free tier)
Audit logging	SQLite
Escalation alerts	Slack Incoming Webhooks
Language	Python 3.11+

Security

All API keys externalised to environment variables, never committed to GitHub. An .env.example is provided as a template.

Externalised config

Model names and temperatures are configurable via environment variables — model swaps require no code changes.

Gayatri Anne

AI & Cloud Architect | 18+ Years Enterprise IT

I build AI-powered automation systems that eliminate manual work from business processes, combining agentic workflows, large language models and cloud integration to deliver production-ready solutions.

From multi-agent orchestration and RAG-based knowledge assistants to intelligent email triage and order processing agents, I design systems that connect AI to the tools businesses already use — cloud platforms, CRM, communication APIs and data stores — with observability and human oversight built in.

Certifications

- Azure Solutions Architect Expert
- Azure AI Engineer Associate
- Azure Administrator
- Azure AI Fundamentals
- Azure Fundamentals
- TOGAF Foundation
- Python (PCAP)